MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – C)** | **Time:** 1 Hour

Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 30

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. If the distance between the points $A(4, p)$ and $B(1, 0)$ is 5 units, then the value of p is: [1]

(A) 4 only **(B)** ± 4 **(C)** -4 only **(D)** 0

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) ± 4

Given: $A(4, p)$, $B(1, 0)$, and distance $AB = 5$ units.

We know that the distance between (x_1, y_1) and (x_2, y_2) is given by:

Formula Used: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Substituting $AB = 5$, $x_1 = 4$, $y_1 = p$, $x_2 = 1$, and $y_2 = 0$:

$$5 = \sqrt{(1-4)^2 + (0-p)^2} = \sqrt{(-3)^2 + (-p)^2} = \sqrt{9 + p^2}$$

Squaring both sides:

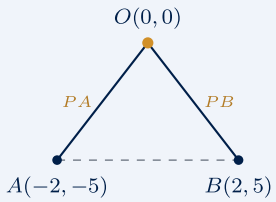
$$25 = 9 + p^2 \implies p^2 = 25 - 9 = 16 \implies \boxed{p = \pm 4}$$

Q2. The point which lies on the perpendicular bisector of the line segment joining the points $A(-2, -5)$ and $B(2, 5)$ is: [1]

(A) $(0, 0)$ **(B)** $(0, 2)$ **(C)** $(2, 0)$ **(D)** $(-2, 0)$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(A)** $(0, 0)$



Given: Points $A(-2, -5)$ and $B(2, 5)$. Every point on the perpendicular bisector of AB is equidistant from A and B, and passes through the midpoint of AB.

We apply the midpoint formula:

Formula Used: Midpoint $M(x, y) = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$

Finding the midpoint M of AB:

$$M = \left(\frac{-2+2}{2}, \frac{-5+5}{2}\right) = \left(\frac{0}{2}, \frac{0}{2}\right) = \boxed{(\mathbf{0}, \mathbf{0})}$$

Moreover, for $O(0, 0)$: $OA = \sqrt{(-2)^2 + (-5)^2} = \sqrt{29} = OB$. Thus, $(0, 0)$ lies on the perpendicular bisector.

Q3. The distance between the points $(a\cos\theta + b\sin\theta,\ 0)$ and $(0,\ a\sin\theta - b\cos\theta)$ is: [1]

(A) $a^2 + b^2$ **(B)** $a + b$ **(C)** $a^2 - b^2$ **(D)** $\sqrt{a^2 + b^2}$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(D)** $\sqrt{a^2 + b^2}$

Given: Points $P(a\cos\theta + b\sin\theta,\ 0)$ and $Q(0,\ a\sin\theta - b\cos\theta)$.

We know the distance formula:

Formula Used: $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Substituting the coordinates and expanding the squares:

$$\begin{aligned} PQ &= \sqrt{(0 - (a\cos\theta + b\sin\theta))^2 + ((a\sin\theta - b\cos\theta) - 0)^2} \\ &= \sqrt{(a\cos\theta + b\sin\theta)^2 + (a\sin\theta - b\cos\theta)^2} \\ &= \sqrt{(a^2\cos^2\theta + 2ab\sin\theta\cos\theta + b^2\sin^2\theta) + (a^2\sin^2\theta - 2ab\sin\theta\cos\theta + b^2\cos^2\theta)} \\ &= \sqrt{a^2(\cos^2\theta + \sin^2\theta) + b^2(\sin^2\theta + \cos^2\theta)} = \boxed{\sqrt{\mathbf{a^2 + b^2}}} \end{aligned}$$

(using the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$).

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Q4. **Assertion (A):** The point $(0, 4)$ lies on the y-axis.
Reason (R): The x-coordinate of any point lying on the y-axis is always zero. [1]

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A)
Evaluating Reason (R): Any point on the y-axis is at a perpendicular distance of 0 units from the y-axis, so its abscissa (x-coordinate) is always 0 (general form $(0, y)$). Thus, Reason (R) is **true**.
Evaluating Assertion (A): In the point $(0, 4)$, we have $x = 0$ and $y = 4$. By Reason (R), since $x = 0$, the point $(0, 4)$ lies on the y-axis (at 4 units above the origin).
Formula Used: Point on the y-axis $\iff x = 0$ (form $(0, y)$)
Thus, both (A) and (R) are true and (R) is the correct explanation of (A).

Q5. The distance between the points $A(0, 6)$ and $B(0, -2)$ is: [1]

(A) 6 **(B)** 8 **(C)** 4 **(D)** 2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 8
Given: $A(0, 6)$ and $B(0, -2)$.
We know the distance formula:
Formula Used: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Substituting the coordinates:

$$AB = \sqrt{(0-0)^2 + (-2-6)^2} = \sqrt{0^2 + (-8)^2} = \sqrt{64} = \boxed{\textbf{8 units}}$$

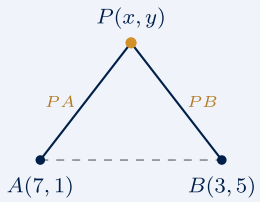
SECTION B: Short Answer Type Questions MATHS MANTHAN | (Printed Q6, Q8–Q10: 8 Marks)

Q6. Prove that abscissa of a point P which is equidistant from points with coordinates $A(7, 1)$ and $B(3, 5)$ is 2 more than its ordinate. [2]

— OR —

Find the coordinates of a point on the line $x + y = 5$ which is equidistant from $(6, 4)$ and $(5, 2)$. [2]

▶ SOLUTION (OPTION I): [Total: 2 Marks]



Step 1: Equating Distances $PA = PB$ [0.5 Mark]
Let the point be $P(x, y)$. Given $PA = PB \implies PA^2 = PB^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Step 2: Expansion and Relation Proof [1.5 Marks]

$$(x-7)^2 + (y-1)^2 = (x-3)^2 + (y-5)^2$$

$$x^2 - 14x + 49 + y^2 - 2y + 1 = x^2 - 6x + 9 + y^2 - 10y + 25$$

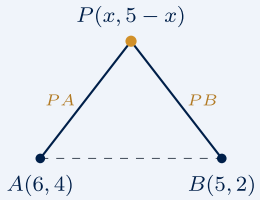
Cancelling $x^2 + y^2$ from both sides:

$$-14x - 2y + 50 = -6x - 10y + 34 \implies -14x + 6x = -10y + 2y + 34 - 50$$

$$-8x = -8y - 16 \implies x = y + 2 \quad \text{(dividing by } -8\text{)}$$

Hence, the abscissa (x) is 2 more than the ordinate (y): $\boxed{\mathbf{x = y + 2}\ \text{(Proved)}}$.

▶ SOLUTION (OPTION II): [OR: 2 Marks]



Step 1: Equating Distances $PA = PB$ [1 Mark]
Let $P(x, y)$ lie on $x + y = 5$ and be equidistant from $A(6, 4)$ and $B(5, 2)$, so $PA^2 = PB^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Using $PA^2 = PB^2$:

$$(x-6)^2 + (y-4)^2 = (x-5)^2 + (y-2)^2$$

$$x^2 - 12x + 36 + y^2 - 8y + 16 = x^2 - 10x + 25 + y^2 - 4y + 4$$

$$-12x - 8y + 52 = -10x - 4y + 29 \implies 2x + 4y = 23$$

Step 2: Solving Linear System [1 Mark]
From $x + y = 5$, we get $x = 5 - y$. Substituting into $2x + 4y = 23$:

$$2(5 - y) + 4y = 23 \implies 10 - 2y + 4y = 23 \implies 2y = 13 \implies y = \frac{13}{2}$$

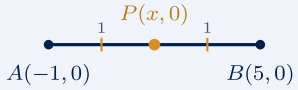
Then $x = 5 - \dfrac{13}{2} = \dfrac{10 - 13}{2} = -\dfrac{3}{2}$.

$$\boxed{\text{Required point } = \left(-\frac{\mathbf{3}}{\mathbf{2}}, \frac{\mathbf{13}}{\mathbf{2}}\right)}$$

Note: *In the printed source question paper (Set – C), Question 7 is absent (the paper transitions directly from Q6 OR on Page 1 to Section B Continued: Questions 8 to 10 on Page 2). Question numbering below preserves the source paper verbatim.*

Q8. Find the coordinates of a point P on the x-axis equidistant from the points $A(-1,0)$ and $B(5,0)$, using the distance formula. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Setup Using Distance Formula [0.5 Mark]
Given: $A(-1,0)$, $B(5,0)$. Since P lies on the x-axis, let P be $(x,0)$. Given $PA = PB \implies PA^2 = PB^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Step 2: Solving for x [1.5 Marks]

$$(x-(-1))^2 + (0-0)^2 = (x-5)^2 + (0-0)^2$$

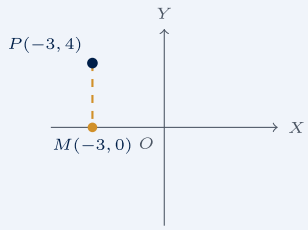
$$(x+1)^2 = (x-5)^2 \implies x^2 + 2x + 1 = x^2 - 10x + 25$$

$$2x + 10x = 25 - 1 \implies 12x = 24 \implies x = \frac{24}{12} = 2$$

$$\boxed{\text{Coordinates of } P = (\mathbf{2}, \mathbf{0})}$$

Q9. Write the distance of the point $(-3,4)$ from the x-axis. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Identifying Perpendicular Foot on x-Axis [1 Mark]
Given: Point $P(-3,4)$. The foot of the perpendicular dropped from $P(-3,4)$ onto the x-axis is $M(-3,0)$.

We know the distance formula:

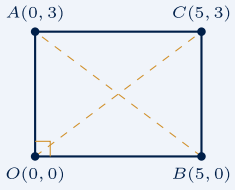
Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ or Distance from x-axis $= |y|$

Step 2: Distance Calculation [1 Mark]

$$PM = \sqrt{(-3-(-3))^2 + (0-4)^2} = \sqrt{0^2 + (-4)^2} = \sqrt{16} = \boxed{\mathbf{4}\text{ units}}$$

Q10. If $AOBC$ is a rectangle whose vertices are $A(0,3)$, $O(0,0)$, $B(5,0)$ and $C(5,3)$, then find the length of its diagonal. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Identifying a Diagonal [0.5 Mark]
Given: Vertices of rectangle $AOBC$ are $A(0,3)$, $O(0,0)$, $B(5,0)$, and $C(5,3)$. Its two equal diagonals are AB and OC.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

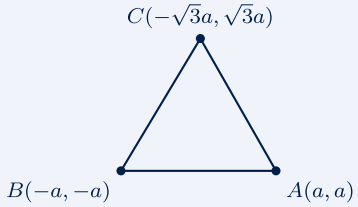
Step 2: Calculating Diagonal Length [1.5 Marks]
Using the distance formula on $A(0,3)$ and $B(5,0)$:

$$AB = \sqrt{(5-0)^2 + (0-3)^2} = \sqrt{5^2 + (-3)^2} = \sqrt{25+9} = \boxed{\sqrt{\mathbf{34}}\text{ units}}$$

(Equivalently, diagonal $OC = \sqrt{(5-0)^2 + (3-0)^2} = \sqrt{34}$ units.)

Q11. Show that the points (a, a), $(-a, -a)$ and $(-\sqrt{3}a,\ \sqrt{3}a)$ are the vertices of an equilateral triangle. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Declaring Vertices and Distance Formula [1 Mark]
Let the given points (with $a \neq 0$) be $A(a, a)$, $B(-a, -a)$, and $C(-\sqrt{3}a,\ \sqrt{3}a)$.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 2: Calculating All Three Side Lengths [3 Marks]

1. Length of side AB:

$$AB = \sqrt{(-a - a)^2 + (-a - a)^2} = \sqrt{(-2a)^2 + (-2a)^2} = \sqrt{4a^2 + 4a^2} = \sqrt{8a^2} = \mathbf{2\sqrt{2}|a|}$$

2. Length of side BC:

$$BC = \sqrt{(-\sqrt{3}a - (-a))^2 + (\sqrt{3}a - (-a))^2} = \sqrt{a^2(1 - \sqrt{3})^2 + a^2(\sqrt{3} + 1)^2}$$
$$= \sqrt{a^2\left[(1 - 2\sqrt{3} + 3) + (3 + 2\sqrt{3} + 1)\right]} = \sqrt{a^2(4 + 4)} = \sqrt{8a^2} = \mathbf{2\sqrt{2}|a|}$$

3. Length of side AC:

$$AC = \sqrt{(-\sqrt{3}a - a)^2 + (\sqrt{3}a - a)^2} = \sqrt{a^2(\sqrt{3} + 1)^2 + a^2(\sqrt{3} - 1)^2}$$
$$= \sqrt{a^2\left[(3 + 2\sqrt{3} + 1) + (3 - 2\sqrt{3} + 1)\right]} = \sqrt{a^2(4 + 4)} = \sqrt{8a^2} = \mathbf{2\sqrt{2}|a|}$$

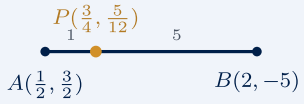
Step 3: Conclusion [1 Mark]
Since all three sides are equal ($AB = BC = AC = 2\sqrt{2}|a|$ for $a \neq 0$):

$\triangle$**ABC** is an equilateral triangle (Hence Proved)

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Q12. Find the ratio in which the point $P\left(\frac{3}{4}, \frac{5}{12}\right)$ divides the line segment joining the points $A\left(\frac{1}{2}, \frac{3}{2}\right)$ and $B(2, -5)$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Step 1: Setting Ratio $k : 1$ and Section Formula [1.5 Marks]
Given: $A\left(\frac{1}{2}, \frac{3}{2}\right)$, $B(2, -5)$, and dividing point $P\left(\frac{3}{4}, \frac{5}{12}\right)$. Let P divide AB in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$

Step 2: Solving Using x-Coordinate [2 Marks]
Substituting $m_1 = k$, $m_2 = 1$, $x_1 = \frac{1}{2}$, $x_2 = 2$, and $x = \frac{3}{4}$:

$$\frac{k(2) + 1\left(\frac{1}{2}\right)}{k + 1} = \frac{3}{4}$$

Cross-multiplying:

$$4\left(2k + \frac{1}{2}\right) = 3(k + 1) \implies 8k + 2 = 3k + 3 \implies 8k - 3k = 3 - 2 \implies 5k = 1 \implies k = \frac{1}{5}$$

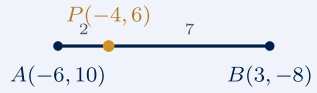
Step 3: Verification Using y-Coordinate [1.5 Marks]
Checking the y-coordinate for $k = \frac{1}{5}$ (ratio $1 : 5$):

$$y = \frac{1(-5) + 5\left(\frac{3}{2}\right)}{1 + 5} = \frac{-5 + \frac{15}{2}}{6} = \frac{\frac{-10+15}{2}}{6} = \frac{5}{12} \quad \text{(Matches given ordinate!)}$$

Required ratio $= \mathbf{1 : 5}$

Q13. In what ratio does the point $P(-4, 6)$ divide the line segment joining the points $A(-6, 10)$ and $B(3, -8)$? **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Step 1: Assuming Ratio $k : 1$ [1.5 Marks]
Given: $A(-6, 10)$, $B(3, -8)$, and $P(-4, 6)$. Let P divide AB internally in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$

Step 2: Equating the x-Coordinate [2 Marks]

$$\frac{k(3) + 1(-6)}{k + 1} = -4 \implies 3k - 6 = -4(k + 1) \implies 3k - 6 = -4k - 4$$

$$3k + 4k = 6 - 4 \implies 7k = 2 \implies k = \frac{2}{7}$$

Step 3: Verification with y-Coordinate [1.5 Marks]
Checking ordinate with $m_1 : m_2 = 2 : 7$:

$$y = \frac{2(-8) + 7(10)}{2 + 7} = \frac{-16 + 70}{9} = \frac{54}{9} = 6 \quad \text{(Matches } y = 6\text{)}$$

Required ratio $= \mathbf{2 : 7}$

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